Which common NIRS variable reflects muscle estimated lactate threshold most closely?

Lixin Wang, Takahiro Yoshikawa, Taketaka Hara, Hayato Nakao, Takashi Suzuki, and Shigeo Fujimoto

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Abstract: Various near-infrared spectroscopy (NIRS) variables have been used to estimate muscle lactate threshold (LT), but no study has determined which common NIRS variable best reflects muscle estimated LT. Establishing the inflection point of 2 regression lines for deoxyhaemoglobin ($\Delta HHb_{i.p.}$), oxyhaemoglobin ($\Delta O_2Hb_{i.p.}$), and tissue oxygenation index ($TOI_{i.p.}$), as well as for blood lactate concentration, we then investigated the relationships between NIRS variables and ventilatory threshold (VT), LT, or maximal tissue hemoglobin index ($nTHI_{max}$) during incremental cycling exercise. $\Delta HHb_{i.p.}$ and $TOI_{i.p.}$ could be determined for all 15 subjects, but $\Delta O_2Hb_{i.p.}$ was determined for only 11 subjects. The mean absolute values for the 2 measurable slopes of the 2 continuous linear regression lines exhibited increased changes in 3 NIRS variables. The workload and VO_2 at $\Delta O_2Hb_{i.p.}$ and $nTHI_{max}$ were greater than those at VT, LT, $\Delta HHb_{i.p.}$, and $TOI_{i.p.}$. For workload and VO_2, $\Delta HHb_{i.p.}$ was correlated with VT and LT, whereas $\Delta O_2Hb_{i.p.}$ was correlated with $nTHI_{max}$, and $TOI_{i.p.}$ with VT and $nTHI_{max}$. These findings indicate that ΔO_2Hb strongly corresponds with local perfusion, and TOI corresponds with both local perfusion and deoxygenation, but that ΔHHb can exactly determine deoxygenation changes and reflect O_2 metabolic dynamics. The finding of strongest correlations between ΔHHb and VT or LT indicates that ΔHHb is the best variable for muscle LT estimation.

Key words: NIRS inflection point, dynamics of O_2Hb, muscle perfusion, ventilatory threshold, lactate threshold.

Résumé : On utilise diverses variables de la spectroscopie dans le proche infrarouge (NIRS) pour estimer le seuil lactique dans le muscle (LT) .Cependant aucune étude n'identifie la meilleure variable à utiliser. Après avoir établi le point d'inflexion des 2 droites de régression de la désoxyhémoglobine (ΔHHb_{ip}), de l'oxyhémoglobine (ΔO_2Hb_{ip}) et de l'index d'oxygénation tissulaire (TOIip) en fonction de l'intensité de travail, nous avons analysé la relation entre les variables de la NIRS et le seuil ventilatoire (VT), le LT et l'index maximal de l'hémoglobine tissulaire ($nTHI_{max}$) au cours d'une épreuve d'intensité progressive sur ergocycle. Nous avons identifié la $\Delta HHbip$ et le TOIip chez 15 sujets et le ΔO_2Hb_{ip} chez seulement 11 sujets. Les moyennes des valeurs absolues des pentes respectives des 2 droites de régression indiquent des augmentations des trois variables de la NIRS. Les valeurs de l'intensité de travail et de la consommation d'oxygène correspondant à ΔO_2Hb_{ip} et $nTHI_{max}$ sont supérieures à celles correspondant à VT, LT, ΔHHb_{ip} et TOI_{ip}. Tant pour l'intensité de travail que pour la consommation d'oxygène, ΔHHb_{ip} est corrélé à VT et à LT tandis que ΔO_2Hb_{ip} est corrélé à $nTHI_{max}$ alors que TOI_{ip} l'est à VT et à nTHImax. Ces observations indiquent que les variations de Δ oxyhémoglobine reflètent la perfusion locale et que TOI reflète la perfusion locale et la désoxygénation; la Δ désoxyhémoglobine (ΔHHb) peut déterminer précisément les variations de désoxygénation et refléter la dynamique de l'oxygène dans le métabolisme. La forte corrélation entre ΔHHb et VT ou LT indique que ΔHHb est la meilleure variable pour estimer le LT musculaire.

Mots clés : point d'inflexion des variables de la NIRS, dynamique de O_2Hb, perfusion musculaire, seuil ventilatoire, seuil lactique.

[Traduit par la Rédaction]

Introduction

Although the capacity for aerobic endurance activity is usually assessed by maximal O_2 uptake ($VO_{2\ max}$), ventilatory threshold (VT) (Beaver et al. 1986), or lactate threshold (LT) (Davis 1985), these are all systemic metabolic variables. Some sports such as cycling, canoe racing, and wheelchair racing are mainly performed with only the lower or upper limbs, and the metabolic dynamics (such as O_2 consumption) and aerobic endurance capacity in local working muscle might not be reflected in systemic metabolic status. Differences in absorption characteristics of oxyhaemoglobin (O_2Hb) and deoxyhaemoglobin (HHb) in the near-infrared spectrum (NIRS) have proven useful for estimating relative changes in local muscle oxygenation and blood volume, as well as the dynamics between local muscle O_2 supply and muscle O_2 consumption in real time during exercise (Belardinelli et al. 1995; Bhambhani et al. 1997; Binzoni et al. 1998; Boushel et al. 2001; Grassi et al. 2003; Quaresima et al. 2003).

A previous study (Kawaguchi et al. 2001) investigated the relationship between the kinetics of muscle oxygenation and

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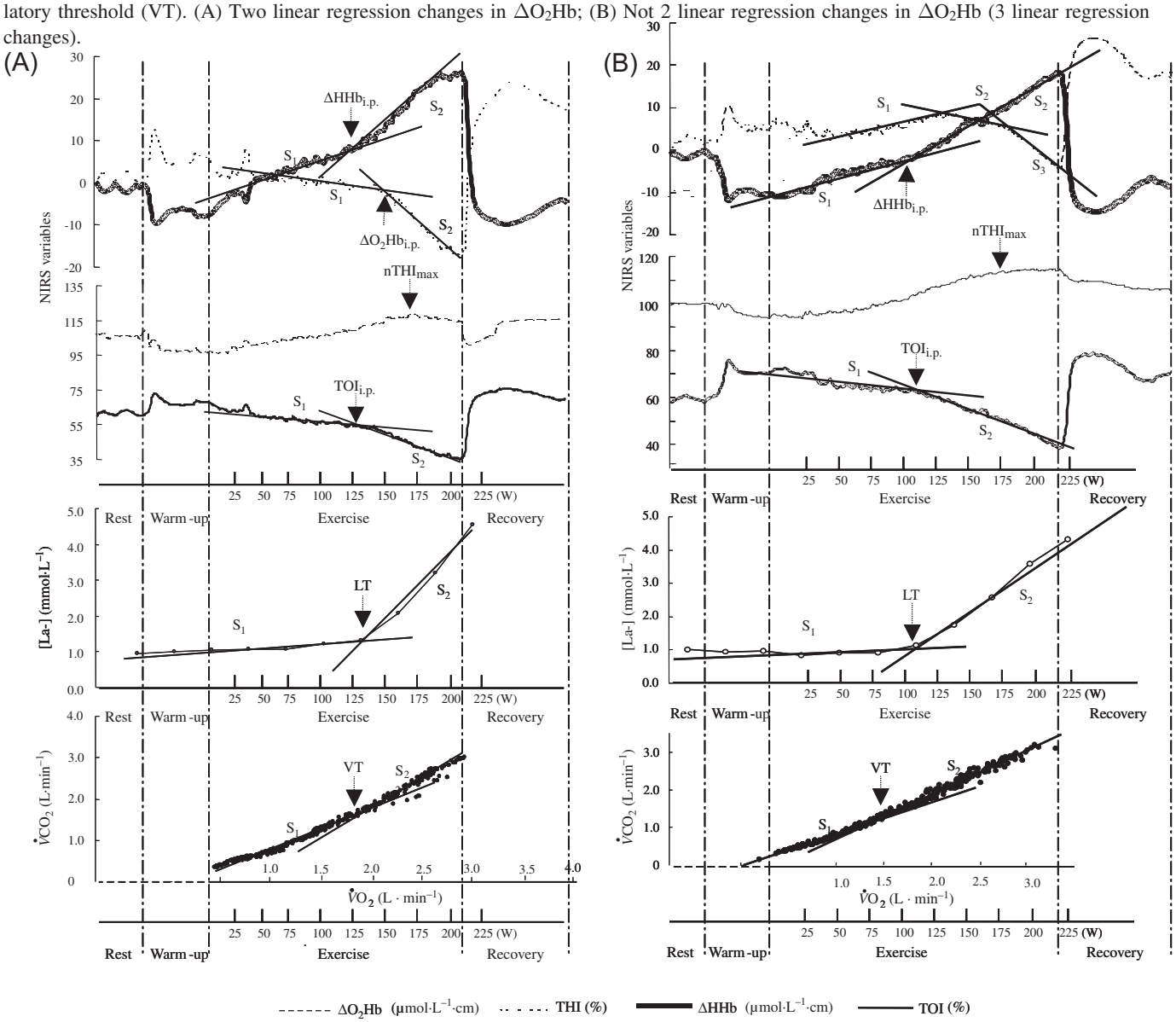
Fig. 1. Trends in 4 NIRS variables (ΔO_2Hb, ΔHHb, TOI, and nTHI) measured from the right vastus lateralis muscle before (rest and warm-up), during, and after incremental cycling exercise in 2 representative subjects by comparison with lactate threshold (LT) and ventilatory threshold (VT). (A) Two linear regression changes in ΔO_2Hb; (B) Not 2 linear regression changes in ΔO_2Hb (3 linear regression changes).



systemic VO_2 in an exercise tolerance test on a bicycle ergometer, and reported that VO_2 exhibited a positive correlation with HHb and a negative correlation with O_2Hb and O_2 saturation. Miura et al. (1998) compared change in O_2Hb and anaerobic threshold (AT) in subjects with different levels of fitness (6 active individuals, 21 sedentary individuals, and 16 patients with heart failure), and found that O_2Hb began to decrease at 42.0%$VO_{2\ max}$ near the AT (47.1% $VO_{2\ max}$) during incremental exercise. They also acknowledged that in some cases the onset of O_2Hb decrease could not be determined due to increases in total Hb, and that when O_2Hb decreases were not clear, inflection points were determined from HHb data. However, they did not further investigate the differences between changes of O_2Hb and HHb. Other studies (Belardinelli et al. 1995; Bhambhani et al. 1997; Grassi et al. 1999) have attempted to determine the AT point using O_2Hb-associated variables. Belardinelli et al. (1995) identified the NIRS AT by introducing the "accelerated fall in tissue O_2 saturation" concept using O_2Hb, but because total Hb is not constant, whether O_2Hb reflects real-time O_2 metabolic kinetics is still unclear (Binzoni et al. 1998). Although Bhambhani et al. (1997) identified the NIRS AT as the time segment at which O_2Hb crossed its baseline value recorded immediately prior to the onset of exercise, we believe that with use of O_2Hb as the oxygenation index, determination of the baseline value itself is not easy because of limits to the sensitivity of NIRS measurement. In addition, Grassi et al. (1999) examined the relationship between muscle oxygenation and LT, and found that the onset of muscle oxygenation was significantly associated with LT during incremental cycling exercise. However, no warm-up exercise was included in their protocol, and their muscle deoxygenation index (ΔO_2Hb – HHb) could also have been affected by the increase in arterial perfusion dur-

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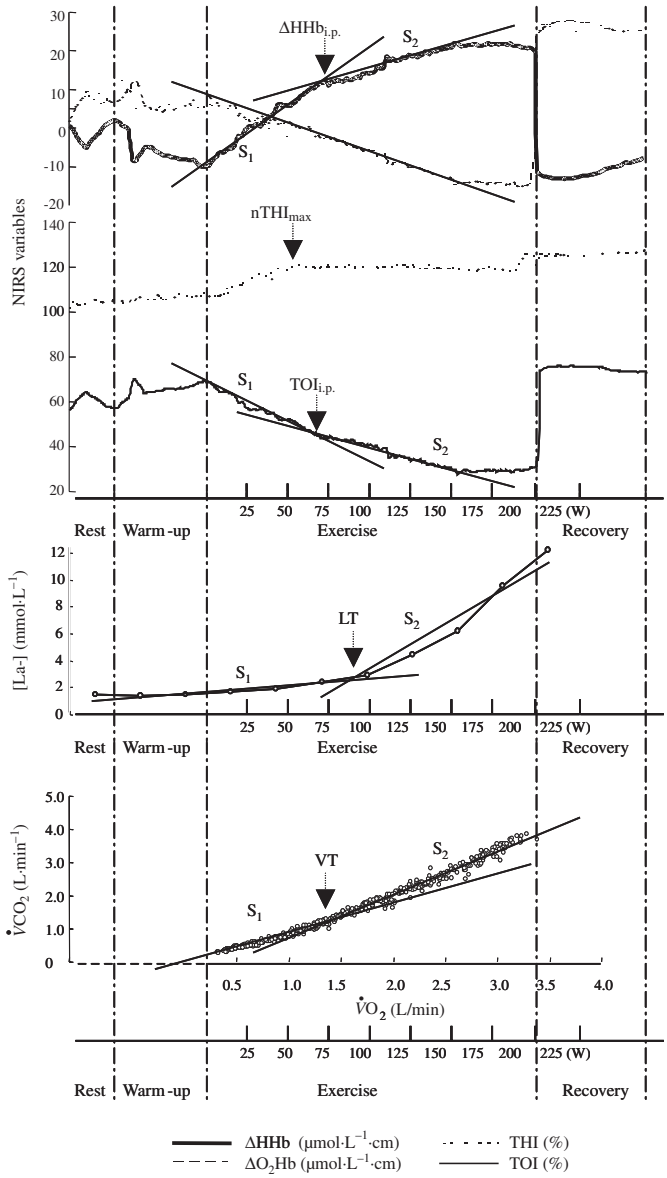
Fig. 2. A typical subject exhibited decreased absolute slopes ($|S_2| < |S_1|$ of ΔHHb and TOI, which was measured from the right vastus lateralis muscle before (rest and warm-up), during, and after incremental cycling exercise, by comparison with lactate threshold (LT) and ventilatory threshold (VT). ΔO_2Hb exhibited a simple linear decrease with 1 slope.



ing incremental exercise. In conclusion, the methodologies of previous studies need to be reviewed, and it is still unclear which NIRS variables best reflect muscle estimated LT.

The purposes of this study were to determine which common NIRS variable most closely reflects muscle estimated LT and to determine the underlying mechanical differences in NIRS variables. The relationships between the changes in characteristics of HHb (ΔHHb), O_2Hb (ΔO_2Hb), and tissue oxygenation index (TOI) variables and maximal normalized tissue hemoglobin index ($nTHI_{max}$), VT, and LT were investigated. Because ΔHHb basically reflects O_2 consumption by local muscle, we hypothesized that ΔHHb is the best NIRS variable for local LT estimation in the main working muscle.

Materials and methods

Participants

Fifteen active, healthy, males (age, 25.3 ± 6.9 y (range = 19–43 y); height, 170.7 ± 4.1 cm; body mass index (BMI), 21.0 ± 1.2 kg·m^{-2}) without competitive cycling experience participated in this study. The subjects consisted of 6 university members, 4 long-distance runners, and 5 soccer players. They were required to avoid high-intensity exercise or training for 3 d before testing and to have a light meal at least 1 h before testing. All subjects were given a bottle of mineral water before exercise. The subjects were informed of the potential risks involved before signing a written voluntary consent form in accordance with the policy statement of the American College of Sports Medicine. Since the voluntary consent form for blood sampling could not be obtained from 6 subjects, blood sampling was actually conducted on 9 subjects. Institutional approval of the study was obtained from the Graduate School of Medicine, Osaka City University, Osaka, Japan.

Experimental design

Testing protocol

Each subject performed incremental exercise to complete exhaustion on a cycling ergometer (Corival, Lode, B.V. Medical Technology, Groningen, Netherlands). Their respiratory gas exchange and NIRS data were recorded before, during, and after exercise. The exercise protocol consisted of 2 min resting, 3 min warm-up exercise (0 W, 60 r/min), incremental cycling exercise, and 6 min recovery. Maximal cycling exercise began at 0 W and 60 r/min and gradually increased at a rate of 15 W/min until exhaustion, which was defined as failing to maintain a pedal rate of 60 r/min. Maximal oxygen consumption ($VO_{2\,max}$) was confirmed by meeting 2 of the following 3 criteria (ACSM 1986): (*i*) leveling off or increase in VO_2 less than 150 mL despite increase in work rate; (*ii*) respiratory exchange ratio (RER) greater than 1.10; and (*iii*) attainment of heart rate higher than 95% of the age-predicted maximum (220 – age). Borg's category-rate scale 10 was used to assess the ratings of perceived exertion (RPE) and to confirm if the subject had reached maximal exercise.

Expired gas analysis

O_2 uptake (VO_2) and CO_2 production (VCO_2) were measured by the breath-by-breath method using a gas analyzer (AE-280, Minato Medical Science, Osaka, Japan), and recorded by computer with a sampling rate of 0.5 Hz. For each episode of exercise, the gas analyzer was calibrated automatically using 2 gases of known concentration, and volume calibration was manually conducted using 2 L air before measurements started being taken. The recorded data were averaged over 9 s intervals using a moving average method. VT was determined by the conventional *V*-slope method, and the respiratory compensation segments of the exercise test were excluded from this analysis (Beaver et al. 1986).

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Table 1. Physiological variables at ventilatory threshold (VT) and maximal cycling exercise (n = 15).

	Power output (W)	VO_2 ($L \cdot min^{-1}$)	VO_2/kg ($mL \cdot min^{-1} \cdot kg^{-1}$)	Heart rate (beats/min)	Minute ventilation ($L \cdot min^{-1}$)	RER	Borg rating of perceived exertion
VT	94±25	1.25±0.29	21.2±3.9	111±12	33.23±5.04	0.91±0.03	4.0±1.0
Maximal value	223±30	2.85±0.42	44.9±6.7	178±12	85.37±15.38	1.16±0.08	10±0

Note: Values are mean ± SD.

Table 2. The individual changes of 2 linear regression slopes ($\mu mol \cdot L^{-1} \cdot cm \cdot W^{-1}$) for ΔHHb, TOI, and ΔO_2Hb during incremental cycling exercise.

	ΔHHb		TOI		ΔO2Hb	
Subjects	S_1	S_2	S_1	S_2	S_1	S_2
1	0.176	0.212	–0.128	–0.196	–0.012	–0.172
2	0.108	0.252	–0.088	–0.284	–0.020	–0.244
3	0.092	0.216	–0.076	–0.220	–0.064	–0.184
4	0.128	0.140	–0.124	–0.152	–0.052	–0.144
5	0.052	0.196	–0.044	–0.104	–0.036	–0.212
6	0.136	0.140	–0.001	–0.004	–0.132	–0.188
7	0.224	0.276	–0.080	–0.132	–0.068	–0.188
8	0.080	0.180	–0.076	–0.272	–0.020	–0.240
9	0.164	0.228	–0.001	–0.002	–0.028	–0.032
10	0.020	0.064	–0.060	–0.168	–0.044	–0.168
11	*0.240*	*0.112*	*–0.264*	*–0.168*	–0.100	–0.116
12	*0.116*	*0.060*	*–0.132*	*–0.064*	—	—
13	0.128	0.184	–0.001	–0.002	—	—
14	0.120	0.292	–0.100	–0.280	—	—
15	0.068	0.164	–0.048	–0.192	—	—
Means (SD)	0.124 (0.060)	0.181(0.070)*	–0.082 (0.067)	–0.149 (0.098)*	–0.052 (0.037)	–0.172 (0.060)*

Note: All the 15 subjects exhibited the 2 linear regression slopes in ΔHHb and TOI, but only 11 subjects exhibited the typical changes in ΔO_2Hb. Subjects 11 and 12 showed contrarily decreased slopes (italics).

*$p < 0.05$ compared with the S_1 value.

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NIRS

The NIRS apparatus (Niromonitor NIRO-200: Hamamatsu Photonics. Co. JP) utilized in this study was a 3-wavelength continuous wave system, which uses the modified Beer–Lambert and spatially resolved spectroscopy (SRS) methods simultaneously. ΔO_2Hb and ΔHHb were measured using the differences in absorption characteristics of light at 775, 810, and 850 nm. The slope of the NIR light attenuation versus distance relation is measured at a point distant from the light input, from which the absolute ratio of O_2Hb to total Hb content, and hence average tissue O_2 saturation (TOI), as well as total Hb content (nTHI), can be calculated semi-quantitatively using photon diffusion theory (Shiga et al. 1997). SRS-O_2 represents a dynamic balance between O_2 supply and O_2 consumption in the small vessels such as the capillaries and arteriolar and venular beds (Quaresima et al. 2003). The skin was thoroughly shaved before affixing the probe to decrease interfering noise during measurement. The probe unit, molded in an elastic black elastomer holder, has 2 silicon photodiodes as photodetectors on 1 side and 3 light-emitting diodes on the other side. The probe holder was firmly attached to the skin overlying the lower one-third of muscle (for subjects >170 cm tall, it was affixed 12 cm above the knee joint; subjects < 170 cm, 10 cm above the knee joint), parallel to the major axis of the thigh with bi-adhesive tape. To further prevent contamination by ambient light, 3 pieces of black medical tape were placed around the probe holder and the skin. The probe unit was connected to a personal computer for data acquisition, analog-to-digital conversion, and subsequent analysis. The central distance between the light source and the photodiode was 4 cm. The influence of adipose tissue thickness (ATT) on NIRS propagation in leg muscle and on the sensitivity of the NIRS instrument has been investigated by ultrasound examination, which demonstrated that the NIR light penetrates shallow regions of muscle even when ATT is 1.5 cm (Matsushita et al. 1998). In the present study, the average body mass index was 21.1 ± 1.2 $kg \cdot m^{-2}$, lower than the normal value. The ATT of our subjects appeared to permit NIRS penetration sufficient to measure O_2 dynamics (the change or balance between O_2Hb and HHb) within relatively deep muscle. In addition, even though myoglobin (Mb) has the same absorption spectra as Hb, the ratio of [Hb] to [Mb] in human skeletal muscle is > 5 (Mancini 1997), therefore NIRS signals can be considered to derive mainly from Hb.

Determination of NIRS and [La⁻] inflection points

Since the increases or decreases in ΔHHb, TOI, and ΔO_2Hb during incremental exercise visually resemble the changes of plasma lactate concentration ([La⁻]) and V-slope, we attempted to extract the general characteristics of NIRS variables and focused on determination of inflection points at which the slopes of ΔHHb, ΔO_2Hb, and TOI began to change. The changes of slope signify that ΔHHb, ΔO_2Hb,

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Table 3. The mean values of workload and oxygen uptake for ventilatory threshold (VT), lactate threshold (LT), near-infrared spectroscopy inflection points ($\Delta HHb_{i.p.}$, $TOI_{i.p.}$, $\Delta O_2Hb_{i.p.}$), and the maximal normalized tissue hemoglobin index ($nTHI_{max}$).

	Workload	VO_2/kg ($mL \cdot min^{-1} \cdot kg^{-1}$)
VT	105±28a	21.2±3.9c
LT	99±23a	21.4±3.4c
$\Delta HHb_{i.p.}$	106±26a	21.3±3.9c
$TOI_{i.p.}$	108±24a	21.7±4.3c
$\Delta O_2Hb_{i.p.}$	141±25b	26.5±4.6d
$nTHI_{max}$	138±37b	26.7±7.3d

Note: $p < 0.001$ for a vs. b; $p < 0.01$ for c vs. d; $n = 15$, but for $\Delta O_2Hb_{i.p.}$, $n = 11$ and for LT, $n = 8$.

and TOI values have begun to change more quickly or slowly with increased exercise workload.

The inflection points ($\Delta HHb_{i.p.}$, $\Delta O_2Hb_{i.p.}$ and $TOI_{i.p.}$, indicated by arrows in Fig. 1A and 1B and Fig. 2) at which the change in slope occurred were determined by iteratively fitting different combinations of 2 regression lines (slope 1, S_1; slope 2, S_2) to contiguous experimental points obtained during exercise and by determining which combination yielded the lowest sum of squared residuals. The fitting data started from the points (about 2–3 min from the start of incremental exercise) where the values began to exhibit consistent trends with increasing exercise intensity and ended at the peak (ΔHHb) or nadir points (ΔO_2Hb, TOI). Similarly, the [La⁻] inflection point (LT) was determined using the same method for determination of $\Delta HHb_{i.p.}$, $\Delta O_2Hb_{i.p.}$, and $TOI_{i.p.}$.

During measurement, ΔO_2Hb and ΔHHb were expressed in micromoles per litre centimetres ($\mu mol \cdot L^{-1} \cdot cm$) with respect to an initial value (at steady-state at rest) arbitrarily set equal to 0, TOI was expressed in percent (%), and nTHI was normalized to the initial value set equal to 1 (100%). The raw NIRS data were measured with a sampling frequency of 0.5 Hz, and were smoothed by the moving average across 5 data points.

Blood sampling

Two millilitres of brachial venous blood was obtained from 9 subjects, including 4 long-distance runners and 5 soccer players, at rest, during warm-up, and during incremental cycling exercise every 2 min and immediately deproteinized and centrifuged to separate the plasma and blood cells. Owing to technical problems, sampling blood was insufficient for 1 subject. [La⁻] was determined by an enzymatic method (ESAT 6661 Lactate, Eppendorf). Since the occurrence of [La⁻] in arterial blood is consistent with that in muscles (Knuttgen and Saltin 1972), but about 2 min faster than the [La⁻] in venous blood (Yoshida et al. 1982), the measured venous [La⁻] at a certain time point in the present study was actually the [La⁻] value in muscle 2 min earlier.

Statistics

Values are presented as mean ± standard deviation (SD). Equality of variance assumptions were examined in advance by a Bartlett test for each variable. The significance of differences of S_1 and S_2 slopes for the regression lines used to estimate $\Delta HHb_{i.p.}$, $TOI_{i.p.}$, and $\Delta O2Hb_{i.p.}$ was examined using a paired t test. One-way factorial analysis of variance (ANOVA) was used to examine the significance of differences among variables in VT, LT, and NIRS, with Fisher's PLSD test performed for post hoc analysis. Correlations between the variables at $\Delta HHb_{i.p.}$, $TOI_{i.p.}$, $\Delta O_2Hb_{i.p.}$ and VT, LT, and $nTHI_{max}$ were examined by determination of Pearson's correlation coefficients. Findings of $p < 0.05$ were considered significant. Analyses were performed using DA Stats Macintosh and StatView Windows version 5.0 (SAS Institute, Cary, N.C.).

Results

Physiological responses at maximal cycling exercise and VT

The mean values for power output, VO_2, VO_2/kg, heart rate, minute ventilation, RER, and RPE at maximal cycling exercise and VT are shown in Table 1. The mean maximal [La⁻] was 6.3 ± 2.1 $mmol \cdot L^{-1}$.

Changes in NIRS variables

Changes in NIRS variables, LT, and VT in a representative case before, during, and after exercise are shown in Fig. 1A. With increase in workload after the first 2–3 min of incremental exercise, NIRS values began to exhibit consistent changes as follows:

- nTHI increased linearly until reaching its peak ($nTHI_{max}$), and then leveled off to the end of exercise. The individual $nTHI_{max}$ points exhibited large variation.
- ΔHHb began to increase rapidly (S_2) after a slow linear increase (S_1), while ΔO_2Hb and TOI began to decrease rapidly (S_2) after a slow linear decrease (S_1). Visual inspection suggested that the increase/decrease pattern in ΔHHb and TOI appeared to be characterized by 2 regression lines in all 15 subjects, whereas only 11 subjects exhibited the same patterns in ΔO_2Hb and 4 subjects exhibited unclear changes such as initial increase followed by decrease with 3 different visually discriminable slopes (see Fig. 1B, S_1, S_2, and S_3) or simple linear decrease with 1 slope (see Fig. 2).

The individual S_1 and S_2 slopes for the regression lines used to estimate $\Delta HHb_{i.p.}$, $TOI_{i.p.}$, and $\Delta O2Hb_{i.p.}$ are presented in Table 2. The absolute mean S_2 ($|S_2|$) was significantly greater than the absolute mean S_1 ($|S_1|$) in ΔHHb ($p < 0.05$), TOI ($p < 0.05$), and ΔO_2Hb ($p < 0.05$). Increased absolute slope ($|S_2| > |S_1|$) in 13 subjects and decreased absolute slope ($|S_2| < |S_1|$) in 2 subjects (subjects 11 and 12, italics in Table 2) for responses were seen after rather than before $\Delta HHb_{i.p.}$ and $TOI_{i.p.}$. Further, the mean [La⁻] at LT was 1.50±0.41 $mmol \cdot L^{-1}$, with values of 1.35±0.33 $mmol \cdot L^{-1}$ in 6 $|S_2| > |S_1|$ subjects and 1.95 ± 0.35 $mmol \cdot L^{-1}$ in 2 $|S_2| < |S_1|$ subjects.

Comparisons of workload and VO_2 at VT, LT, $\Delta HHb_{i.p.}$, $TOI_{i.p.}$, $\Delta O_2Hb_{i.p.}$, and $nTHI_{max}$

One-way ANOVA revealed significant differences in workload ($p < 0.001$) and VO_2 ($p < 0.01$) at VT, LT, and the 4 NIRS variables. Post hoc testing showed that the workload ($p < 0.001$, in all cases) and VO_2 ($p < 0.01$, in all

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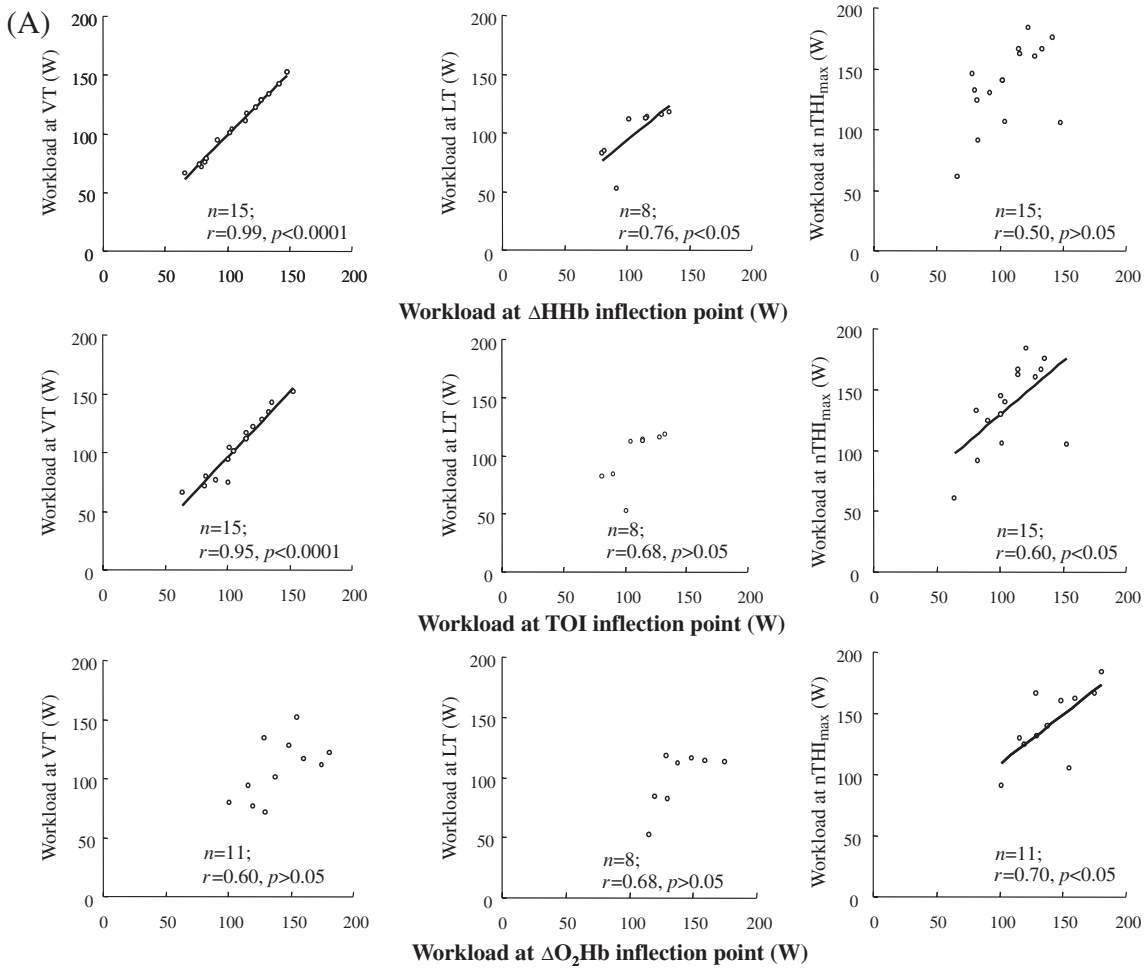
Fig. 3. Correlations of workload (A) and VO_2 (B) values with different oxygenation indexes (inflection point of ΔHHb, upper; TOI, middle; and ΔO_2Hb, lower) and ventilatory threshold (VT), lactate threshold (LT), and maximal normalized tissue hemoglobin index ($nTHI_{max}$). Each symbol represents the value for 1 subject. See text for details.



cases) for $\Delta O_2Hb_{i.p.}$ and $nTHI_{max}$ were significantly greater than for VT, LT, $\Delta HHb_{i.p.}$, and $TOI_{i.p.}$ (Table 3).

Moreover, the workload and VO_2 for $\Delta HHb_{i.p.}$ were significantly correlated with those for VT (r = 0.99, p < 0.0001 and r = 0.97, p < 0.0001, respectively) and LT (r = 0.76, p < 0.05 and r = 0.80, p < 0.05, respectively), but were not correlated with those for $nTHI_{max}$. In contrast, the workload and VO_2 for $\Delta O_2Hb_{i.p.}$ were significantly correlated with those for $nTHI_{max}$ (r = 0.70, p < 0.05 and r = 0.66, p < 0.05, respectively), but were not correlated with those for VT or LT. On the other hand, the workload and VO_2 for $TOI_{i.p.}$ were significantly correlated with those for VT (r = 0.95, p < 0.0001 and r = 0.96, p < 0.0001, respectively) and moderately correlated with those for $nTHI_{max}$ (r = 0.60, p < 0.05 for both), but not with those for LT (Fig. 3A and 3B).

Discussion

The main finding of this study is that, as for the 2 linear regression changes for systemic VCO_2 and VO_2 or [La⁻], comparable changes were observed in ΔHHb, ΔO_2Hb, and TOI variables, with a slower linear change (lower $|S_1|$), followed by a faster linear change (greater $|S_2|$). Inflection points were confirmed in all 15 subjects for ΔHHb or TOI, but in only 11 subjects for ΔO_2Hb (Table 2), and the inflection point of ΔHHb was most strongly correlated with VT and LT.

With increase in exercise intensity, working muscle requires increasing energy consumption with a corresponding increase in O_2 consumption. The increased O_2 requirement may derive from either increase in local muscle perfusion or increase in dissociation of O_2 from O_2Hb, but the rightward shift of the O_2Hb dissociation curve is not large below LT. Accordingly, we infer that the increased O_2 consumption may mainly be supplied by increase in muscle perfusion, indicated by the gradual increases in ΔHHb and nTHI, respectively. Although increase in O_2 consumption or metabolic activity will induce secretion of various vasoactive substances (Laughlin et al. 2004) and arteriolar vasodilation (Clifford and Hellsten 2004), muscle blood flow increases with increase in metabolic activity (Bockman 1983) and reduction in muscle oxygenation (Gonzalez-Alonso et al. 2001), a temporary time delay might occur between increase in metabolic activity and arteriolar vasodilation. Previous studies (Bringard and Perrey 2004; DeLorey et al. 2003) have also shown that a mismatch between local muscle O_2 consumption and perfusion could occur during different types of exercise below AT. We believe that, in

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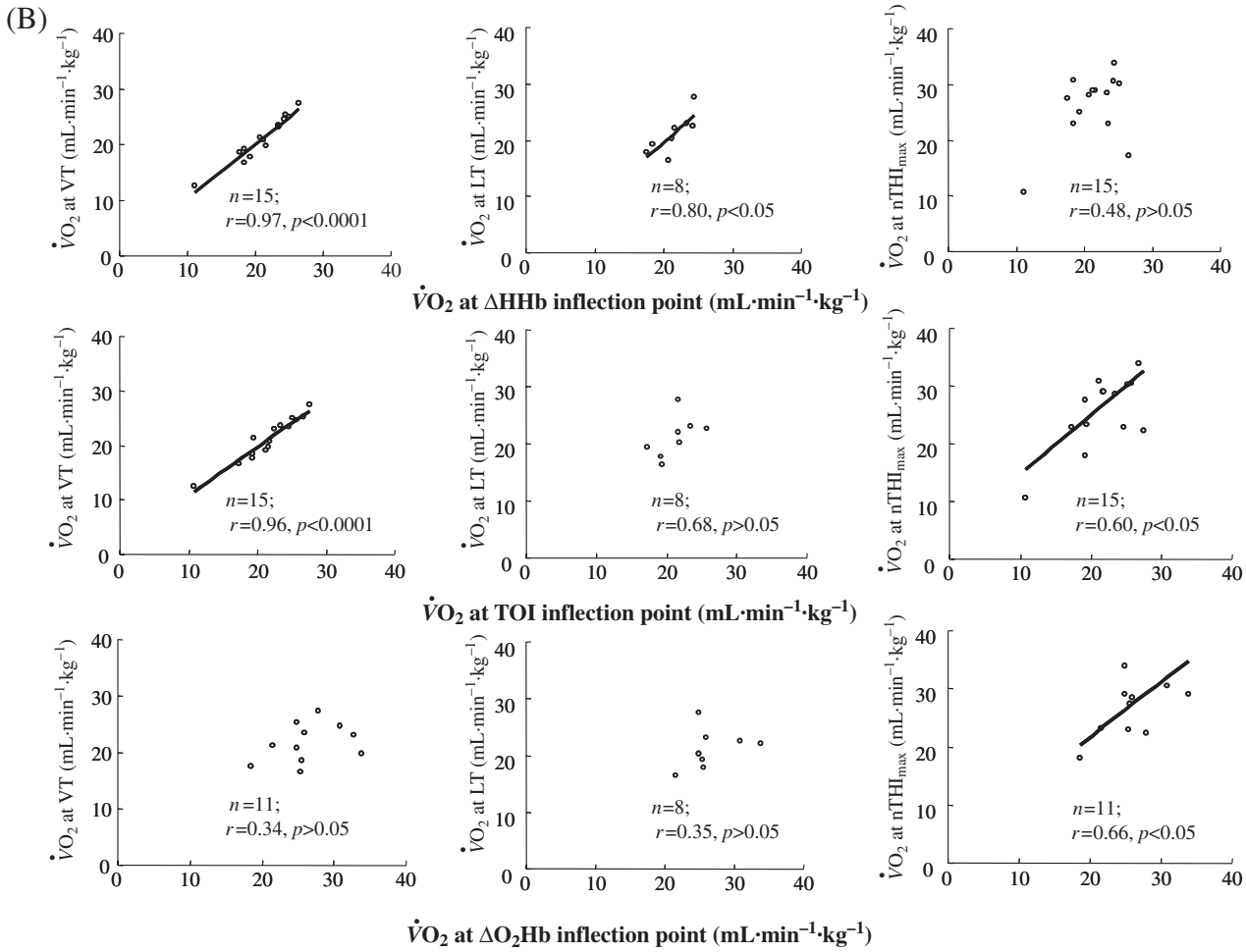
Fig. 3. (*concluded*).



the present study, the occurrence of temporary mismatch between local muscle O_2 consumption and perfusion resulted in a gradual decrease in ΔO_2Hb and TOI in the first half of incremental exercise below AT. Together, these metabolic changes resulted in initially slow linear changes (lower $|S_1|$) in ΔHHb, ΔO_2Hb, and TOI.

Further increase in exercise intensity above AT results in increased anaerobic metabolism and subsequent changes such as accumulation of lactate, arterial carbon dioxide, and hydrogen ions, yielding a shift to the right of the O_2Hb dissociation curve and enhancement of dissociation of O_2 from Hb as a result of the Bohr effect (Beaver et al. 1986; Wasserman et al. 1981). The Bohr effect induced or accelerated the change of O_2Hb to HHb, with rapid increase in ΔHHb and rapid decrease in ΔO_2Hb or TOI (greater $|S_2|$), beginning from the inflection points. The physiological significance of this is maintenance of steady capillary–venular PO_2 level through accelerated muscle deoxygenation. The ΔO_2Hb inflection point signifies that exercise-related vasodilatory capacity within the muscle has reached its maximum.

Continuous assessment of muscle O_2 dynamics during exercise makes NIRS data influenced by the "vascular portion" of measuring location, and NIRS data becomes a weighted average of oxygenation status in arterioles, capillaries, and venules (Quaresima et al. 2003). Although NIRS data reflects ~84% of the muscle microvascular volume from the capillary volume (Poole et al. 1995), increased muscle blood flow and blood volume might shift the weighted rate of NIRS data from venules to arterioles and capillaries (Boushel et al. 2001; McCully and Hamaoka 2000). This could explain why ΔO_2Hb exhibited a continuous increase change despite increase in workload, even after the inflection points in ΔHHb and TOI (S_1 of ΔO_2Hb in Fig. 1B). ΔO_2Hb data can thus be greatly influenced by muscle perfusion during exercise. Our findings showed that the average workload and O_2 uptake at the ΔO_2Hb inflection point were equivalent and significantly associated with the values at nTHImax, but were approximately 30–40 W and 5–6 mL·min^{-1}·kg^{-1} greater than those at the ΔHHb, TOI inflection points, and VT or LT, respectively. This ΔO_2Hb-derived LT (or AT) we obtained differs from that obtained in previous studies (Belardinelli et al. 1995; Bhambhani et al. 1997), which reported that the AT measured from ΔO_2Hb occurred before the VT identified by gas exchange measurements. However, important differences exist among the 3 studies regarding the criteria used to identify ΔO_2Hb-derived LT (or AT), as noted in the Introduction.

Increased muscle oxygenation during exercise was also observed previously (MacDonald et al. 1999), and was not correlated with simultaneously determined O_2Hb saturation in the vein draining from the exercising muscle. We agree

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with Ferrari et al. (1997) and Binzoni et al. (1998) that ΔO_2Hb can only be used as an index of change in Hb saturation when no variations in blood volume (O_2Hb + HHb) occur. In fact, in more recent studies, ΔHHb has increasingly been used as a muscle oxygenation index instead of ΔO_2Hb-associated variables (Grassi et al. 2003; DeLorey et al. 2003). Hence, our findings, combined with those of previous studies, imply that ΔO_2Hb is inappropriate for assessment of O_2 metabolic dynamics in main working muscle.

The workload and O_2 uptake at the ΔHHb inflection point did not differ from those at LT, and were significantly correlated with those at LT. However, no relationship between the TOI inflection point and LT was found. An abrupt decrease in muscle pH could occur, closely related to LT intensity (Systrom et al. 1990), signifying the occurrence of muscle deoxygenation correlated with the occurrence of LT. Our study also found that the TOI inflection point was significantly correlated with muscle perfusion ($nTHI_{max}$). The above findings indicate that, during exercise, TOI can be influenced by either muscle deoxygenation or exercise hyperemia, but that ΔHHb is a consequence of change in O_2 extraction and microvascular deoxygenation and reflects only the balance between O_2 delivery and O_2 utilization in working muscle.

Finally, the linear changes in both ΔHHb and TOI dynamics observed in the present study yielded increased slopes in 13 subjects and decreased slopes in 2 subjects (Table 2). In contrast with the earlier linear decrease in ΔO_2Hb before the occurrence of the ΔHHb inflection point when $|S_2| < |S_1|$ (Fig. 2), ΔO_2Hb slowly increased (Fig. 1B) or remained at a relatively steady state (Fig. 1A) when $|S_2| > |S_1|$. Moreover, when $|S_2| < |S_1|$, muscle perfusion (nTHI) had already peaked before the occurrences of ΔHHb and TOI inflection points, and further increase in it was impossible. When $|S_2| > |S_1|$, however, nTHI peaked after the inflection points in ΔHHb and TOI. These 2 types of slope change in ΔHHb and TOI might have resulted from differences in vasodilatory responses or adaptations to exercise. When $|S_2| > |S_1|$, increasing muscle O_2 consumption before the inflection points of ΔHHb and TOI was mainly supported by increase in blood flow and blood volume through arteriolar vasodilation, and O_2Hb concentration could be maintained or even increased. However, when $|S_2| < |S_1|$, arteriolar vasodilation had peaked before the inflection points of ΔHHb and TOI; the increased O_2 consumption must have been solely derived from O_2Hb dissociation, and thus both ΔHHb and TOI exhibited greater changes in slope in the first half of incremental exercise. In addition, the different changes in slopes in ΔHHb and TOI might also have been influenced by differences in weighting rates of muscle fibers. Since fast-twitch motor units might be dominant in working muscle in $|S_2| < |S_1|$ subjects, anaerobic metabolism might be initialized very early during incremental exercise, with early Bohr effect. This could also be due in part to the higher lactate concentration in $|S_2| < |S_1|$ subjects than in $|S_2| > |S_1|$ subjects at the inflection points of ΔHHb and TOI.

Conclusion

This study attempted to extract the individual inflection points of NIRS variables including ΔHHb, ΔO_2Hb, and TOI, and examined their correlations with nTHI, VT, and LT, to determine which NIRS variable best reflects muscle estimated LT. In the future, larger samples need to be investigated to confirm if the 2 types of slope changes in ΔHHb and TOI exist generally.

The findings we obtained indicate that during incremental exercise, ΔO_2Hb is strongly dependent on change in local muscle perfusion, TOI is influenced by either local muscle perfusion or muscle deoxygenation, but ΔHHb reflects the relationship between local muscle O_2 delivery (and perfusion) and local muscle O_2 utilization (and thus O_2 extraction). The underlying mechanism of ΔHHb occurrence determined its close correlation with VT and LT, and thus ΔHHb is the best variable for estimation of LT in main working muscle.

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References

American College of Sports Medicine. 1986. Guidelines for exercise testing, and prescription. Lea & Febiger, Philadelphia, Pa. pp. 95-96.

Beaver, W.L., Wasserman, K., and Whipp, B.J. 1986. A new method for detecting anaerobic threshold by gas exchange. J. Appl. Physiol. **60**: 2020–2027. PMID:3087938.

Belardinelli, R., Barstow, T.J., Porsasz, J., and Wasserman, K. 1995. Changes in skeletal muscle oxygenation during incremental exercise measured with near infrared spectroscopy. Eur. J. Appl. Physiol. Occup. Physiol. **70**: 487–492.

Bhambhani, Y.N., Buckley, S.M., and Susaki, T. 1997. Detection of ventilatory threshold using near infrared spectroscopy in men and women. Med. Sci. Sports Exerc. **29**: 402–409. PMID:9139181.

Binzoni, T., Quaresima, V., Barattelli, G., Hiltbrand, E., Gurke, L., Terrier, F., et al. 1998. Energy metabolism and interstitial fluid displacement in human gastrocnemius during short ischemic cycles. J. Appl. Physiol. **85**: 1244–1251. PMID:9760312.

Bockman, E.L. 1983. Blood flow and oxygen consumption in active soleus and gracilis muscle in cats. Am. J. Physiol. **244**: H546–H551. PMID:6837755.

Boushel, R., Langberg, H., Olesen, J., Gonzales-Alonzo, J., Bulow, J., and Kjaer, M. 2001. Monitoring tissue oxygen availability with near infrared spectroscopy (NIRS) in health and disease. Scand. J. Med. Sci. Sports, **11**: 213–222. doi:10.1034/j.1600-0838.2001.110404.x. PMID:11476426.

Bringard, A., and Perrey, S. 2004. Influence of repeated isometric contraction on muscle deoxygenation and pulmonary oxygen uptake kinetics in humans. Clin. Physiol. Funct. Imaging, **24**: 229–236. doi:10.1111/j.1475-097X.2004.00554.x. PMID:15233838.

Clifford, P.S., and Hellsten, Y. 2004. Vasodilatory mechanisms in contracting skeletal muscle. J. Appl. Physiol. **97**: 393–403. doi:10.1152/japplphysiol.00179.2004. PMID:15220322.

Davis, J.A. 1985. Anaerobic threshold: review of the concept and directions for future research. Med. Sci. Sports Exerc. **17**: 6–18. PMID:3884961.

DeLorey, D.S., Kowalchuk, J.M., and Paterson, D.H. 2003. Relationship between pulmonary O_2 uptake kinetics and muscle

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deoxygenation during moderate-intensity exercise. J. Appl. Physiol. **95**: 113–120. PMID:12679363.

Ferrari, M., Binzoni, T., and Quaresima, V. 1997. Oxidative metabolism in muscle. Philos. Trans. R. Soc. Lond. B Biol. Sci. **352**: 677–683. PMID:9232855.

Gonzalez-Alonso, J., Richardson, R.S., and Saltin, B. 2001. Exercising skeletal muscle blood flow in humans responds to reduction in arterial oxyhaemoglobin, but not to altered free oxygen. J. Physiol. **530**: 331–341. doi:10.1111/j.1469-7793.2001.0331l.x. PMID:11208980.

Grassi, B., Pogliaghi, S., Rampichini, S., Quaresima, V., Ferrari, M., Marconi, C., and Cerretelli, P. 2003. Muscle oxygenation and pulmonary gas exchange kinetics during cycling exercise on-transitions in humans. J. Appl. Physiol. **95**: 149–158. PMID:12611769.

Grassi, B., Quaresima, V., Marconi, C., Ferrari, M., and Cerretelli, P. 1999. Blood lactate accumulation and muscle deoxygenation during incremental exercise. J. Appl. Physiol. **87**: 348–355. PMID:10409594.

Kawaguchi, K., Tabusadani, M., Sekikawa, Y., Hayashi, Y., and Onari, K. 2001. Do the kinetics of peripheral muscle oxygenation reflect systemic oxygen intake? Eur. J. Appl. Physiol. **84**: 158–161. doi:10.1007/s004210000352. PMID:11394247.

Knuttgen, H.G., and Saltin, B. 1972. Muscle metabolites and oxygen uptake in short-term submaximal exercise in man. J. Appl. Physiol. **32**: 690–694. PMID:4556852.

Laughlin, M.H., Woodman, C.R., Schrage, W.G., Gute, D., and Price, E.M. 2004. Interval sprint training enhances endothelial function and eNOS content in some arteries that perfuse white gastrocnemius muscle. J. Appl. Physiol. **96**: 233–244. PMID:12923113.

MacDonald, M.J., Tarnopolsky, M.A., Green, H.E., and Hughson, R.L. 1999. Comparison of femoral blood gases and muscle near-infrared spectroscopy at exercise onset in humans. J. Appl. Physiol. **86**: 687–693. PMID:9931209.

Mancini, D. 1997. Application of near infrared spectroscopy to the evaluation of exercise performance and limitations in patients with heart failure. J. Biomed. Opt. **2**: 22–30. doi:10.1117/12.263747.

Matsushita, K., Homma, S., and Okada, E. 1998. Influence of adipose tissue on muscle oxygenation measurement with NIRS instrument. Proc. Soc. Photo-Opt. Instrum. Eng. **3194**: 116–120.

McCully, K.K., and Hamaoka, T. 2000. Near-infrared spectroscopy: what can it tell us about oxygen saturation in skeletal muscle? Exerc. Sport Sci. Rev. **28**: 123–127. PMID:10916704.

Miura, T., Takeuchi, T., Sato, H., Nishioka, N., Tarakado, S., Fujieda, Y., and Ibukiyama, C. 1998. Skeletal muscle deoxygenation during exercise assessed by near-infrared spectroscopy and its relation to expired gas analysis parameters. Jpn. Circ. J. **62**: 649–657. doi:10.1253/jcj.62.649. PMID:9766702.

Poole, D.C., Wagner, P.D., and Wilson, D.F. 1995. Diaphragm microvascular plasma PO_2 measured in vivo. J. Appl. Physiol. **79**: 2050–2057.

Quaresima, V., Lepanto, R., and Ferrari, M. 2003. The use of near infrared spectroscopy in sports medicine. J. Sports Med. Phys. Fitness, **43**: 1–13. PMID:12629456.

Shiga, T., Yamamoto, K., Tanabe, K., Nakase, Y., and Chance, B. 1997. Study of an algorithm based on model experiments and diffusion theory for a portable tissue oximeter. J. Biomed. Opt. **2**: 154–161. doi:10.1117/12.268963.

Systrom, D.M., Kanarek, D.J., Kohler, S.J., and Kazemi, H. 1990. 31P nuclear magnetic resonance spectroscopy study of the anaerobic threshold in humans. J. Appl. Physiol. **68**: 2060–2066. PMID:2361908.

Wasserman, K., Whipp, B.J., and Davis, J.A. 1981. Respiratory physiology of exercise: metabolism, gas exchange, and ventilatory control. Int. Rev. Physiol. **23**: 149–211. PMID:6114073.

Yoshida, T., Takeuchi, N., and Suda, Y. 1982. Arterial versus venous blood lactate increase in the forearm during incremental bicycle exercise. Eur. J. Appl. Physiol. **50**: 87–93. doi:10.1007/BF00952247.

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